

BY 519: Protein Sciences in Therapeutics

Course Code	BY 519
Course Name	Protein Sciences in Therapeutics
L-T-P-C	3-0-0-3
Prerequisites	IC136, or consent of faculty
Intended for	B. Tech. 3rd and 4th year, MS/M. Tech., Ph.D.
Distribution	Core for M. Tech. Biotechnology, elective for others
Approval	9th Senate (May 2016)

Course Contents

- **Introduction**
Introduction and basic concepts of proteins. Protein structure and function. [2 Lectures]
- **Intrinsically Disordered Proteins**
Sequence composition of IDPs, distribution of IDPs in nature and their physiological roles, intrinsically disordered regions, fuzzy complexes, designed linkers, folding and binding mechanisms of IDPs. Protein disorder in signaling and disease in human and plants. [6 Lectures]
- **Thermodynamics for therapeutics**
A brief introduction and thermodynamic principles. Gibbs free energy, thermochemistry and calorimetry. Protein folding theories and structural transitions in polypeptides. Thermodynamic characterization of therapeutic proteins for highest stability and activities. Thermodynamic basis of protein-protein interaction inhibition. [12 Lectures]
- **Protein Engineering – Basic Principles and Rationale**
Identification of putative enzymes in sequence databases. Enzymes, catalysis and kinetics, factors influencing the speed of enzymatic reaction. Enzyme applications, targets of protein engineering, protein engineering approaches, advantages and limitations. Successful stories of application of protein engineering to improve enzyme catalytic efficiency, enzyme stability and folding. [8 Lectures]
- **Module V**
Therapeutic potentials of proteins with specific examples including insulin, anticoagulants, blood substitutes and vaccines. Sequence composition and heteromorphous pairs of proteins. [6 Lectures]
- **Module VI**
Protein misfolding and amyloid diseases: Alzheimer's disease, Parkinson's disease. Signalling involved in misfolding diseases. Transthyretin as amyloid diseases. [8 Lectures]

Text and Reference Books

1. Donald Voet, Charlotte W. Pratt, Judith G. Voet, **Principles of Biochemistry**, 4th Edition, Wiley, 2012.
2. David L Nelson, Michael M Cox, Albert L Lehninger, **Lehninger Principles of Biochemistry**, 6th edition, W.H. Freeman, 2013.
3. Strachan, Tom, and Andrew P. Read/ Irwin H. Segel, **Biochemical calculations: how to solve mathematical problems in general biochemistry**, 2nd Edition, Wiley, 1976.
4. T Palmer, P L Bonner, **Enzymes, Biochemistry, Biotechnology, Clinical Chemistry**, 2nd edition, Woodhead Publishing, 2007.
5. Peter Tompa, Alan Fersht, **Structure and Function of Intrinsically Disordered Proteins**, CRC Press, 2009.
6. David Sheehan, **Physical Biochemistry: Principles and Applications**, 2nd Edition, Wiley, 2009.
7. Several recent papers from peer reviewed journals like Nature, Science, Molecular Therapy, PNAS, Biochemistry, JBC etc.

Specialisation Laboratory courses

The experiments proposed are listed against each theme of specialization electives. Each student will perform experiments from either “Systems Biology (ML1)” theme or “Medical and Nano-biotechnology (ML2)” theme. The proposed practicals will be finalized based on standardization and infrastructure development. Efforts are made to match the laboratory exercises with the special elective course components.